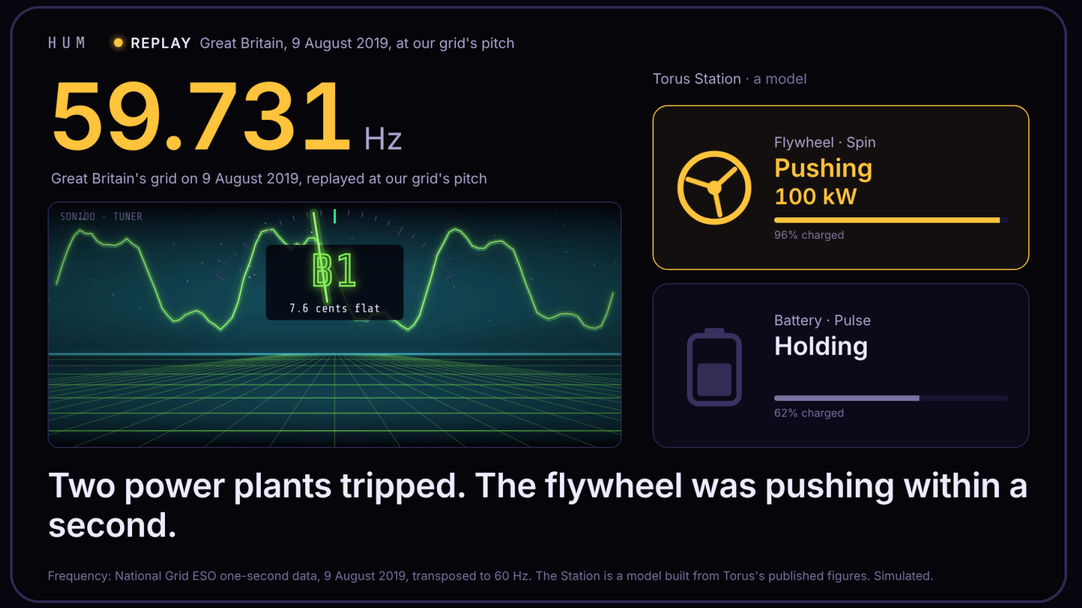
Hard Tech Week 2026 · Torus bounty · Nucleus × Startup State open call

HUM hears the power grid through a guitar, and a model of a Torus Station answers it.

The software that decides is written in Mentl, a programming language I built. Its compiler proves that this software can't reach the internet while it decides.

Morgan Espitia, Salt Lake City. I have built Mentl since March and my audio library, sonido, since January.



Everything on the screens that follow is live in this room, except the replayed 2019 event.

ampactor.dev/hum

Data: grid frequency, NESO open data at 1 s, 9 Aug 2019, moved to 60 Hz · rush hour, EIA-930 hourly demand, PacifiCorp East, 19–26 Oct 2026 · Station: a simulated model of Torus's published figures

How to read this demo

The power grid's frequency shows whether it has enough power. A guitar can hear it.

The grid	The whole Western grid, from Alberta to Baja, runs at one frequency: 60 hertz. When demand outruns supply, the frequency falls.
The ear	Every power adapter hums at that frequency. A guitar pickup resting on one hears it, and my audio library, sonido, measures it to a thousandth of a hertz.
The Station	A model of a Torus Station: a flywheel for the first seconds (100 kW for six minutes) and a battery for the hours (300 kW for four hours). Simulated.
The software	Decides what the Station does. It is written in Mentl, a language I built, whose compiler proves the decision can't reach the internet.
Next	The grid, live in this room. Then tomorrow's rush hour, a real grid failure from 2019, and a blackout.

Station figures: Torus's published specifications, modeled and simulated. Frequency: read live from this room's wiring.

AI now helps write the software that runs machines. No mainstream language can prove what that software will never do.

Batteries, flywheels and flight controls run on code that people test and review, and then trust.

Mentl is a programming language whose compiler proves it.

Mentl proves what software can never do.

You write a rule into the code, such as "this never touches the internet" or "this never writes a file".

The compiler checks the rule through every function the code calls. **Break it, and the program won't build.**

Seven months and 2,600 commits since March. The compiler is written in Mentl and compiles itself. Open source: github.com/ampactor-labs/mentl



Counts: the git history of github.com/ampactor-labs/mentl.

What Mentl code looks like

Five operators draw the shape of a program. The compiler checks each rule along that shape.

`fn read(hz) = hz |> fast`	`|>`	pipe: one stage feeds the next
`fn fork(hz) = (hz) <| (fast, slow)`	`<|`	split: one value goes to two branches
`fn pair(a, b) = (a |> fast) >< (b |> slow)`	`><`	parallel: two branches run side by side
`fn trend(hz) = ((prev) => prev + hz) <~ Delay(1)`	`<~`	feedback: each step sees the step before
`fn act(hz) with !Network = spin(hz) ~> flywheel`	`~>`	handler: names who answers the request

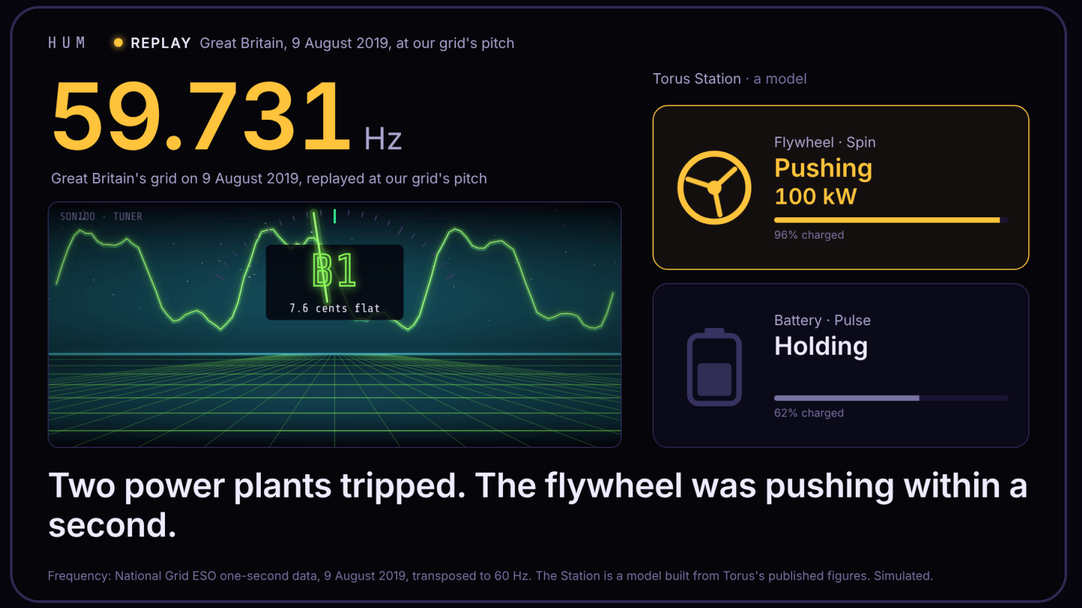
The last line carries a rule, `with !Network`: if anything `act` calls could reach the network, the compiler refuses to build it. Four of the operators come from Faust, the audio language; ~> is Mentl's own.

These lines pass mentl check as written, 28 Oct 2026. The language: github.com/ampactor-labs/mentl

Show it working

Built this week: software that measures the power grid from a wall's wiring.

- **A guitar pickup picks up the grid's hum** in any room, and my audio library, sonido, measures its frequency to a thousandth of a hertz.
- **A model of a Torus Station decides** what to do, typically in under a millisecond, with software written in Mentl.
- **That software is proven** unable to reach the internet while it decides.



Before Monday 4 PM: the language, 2,610 commits since March 15, and sonido, 461 commits since January 25. Since then: everything on this slide.

ampactor.dev/hum

Counts: each repository's git history, printed in pitch/ledger.txt. Picture: the 2019 replay, NESO open data at one second.

Who needs it in Utah

TORUS · SALT LAKE CITY

Builds flywheel-and-battery Stations that steady the grid. The software inside each Station is where a proof would matter first. Torus judges this demo today.

HILL AFB · 309TH SOFTWARE ENGINEERING GROUP

Maintains flight software for the F-16, the F-35 and the missile fleet, where every claim about the software needs evidence.

First conversation I'm asking for: Torus engineering, this month.
First hires: a controls engineer and a compiler engineer, in Salt Lake City.

What $10,000 unlocks

$10,000 puts a proven controller on a real machine in Utah within ninety days.

Day 30 Form the Utah company. Run the controller on a battery inverter on my bench.

Day 60 Run it beside a real battery at one Utah site.

Day 90 A pilot agreement with that site, and an SBIR proposal with Nucleus Grow.

$7,000 for the inverter and battery bench. **$3,000** to form the company, register for federal grants and install at the site.

Three asks. A Utah operator who will host it. An introduction to Hill AFB's 309th. A co-founder who knows power electronics.

Thank you, Torus and Nucleus

Replay this demo on your phone.



`ampactor.dev/hum`

Morgan Espitia · sonofagoose@pm.me

The demo: github.com/ampactor-labs/hum
The language: github.com/ampactor-labs/mentl

Three asks: a Utah operator who will host the pilot, an introduction to Hill AFB's 309th, and a co-founder who knows power electronics.